

VHGRAD

Development of a Didactic Deep Learning Framework from
Scratch

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Technical Progression of the VHgrad Framework

1. Foundations of Neural Computation

Fundamental concepts underlying modern neural networks.

2. Building the First Components

Single neurons, multi-neuron layers, tensors and matrix operations with NumPy.

3. Structured Layer Design

Encapsulation of dense layers into reusable classes with organized data handling.

4. Non-Linearity through Activation

Step, linear, sigmoid, ReLU and Softmax activation functions and their roles.

5. Measuring Network Error

Cross-entropy loss formulation and accuracy metrics for classification.

6. Parameter Optimization

First principles of gradient-based learning and iterative parameter updates.

7. Calculus of Learning

Numerical and analytical derivatives, partial derivatives and the chain rule.

8. Gradient Flow through the Network

Full backward pass with analytically derived gradients across layers and losses.

9. Advanced Optimization Methods

SGD, learning rate scheduling, momentum and adaptive optimizers including Adam.

10. Controlling Generalization

Weight regularization, dropout and validation strategies to reduce overfitting.

11. Extending to Other Task Types

Binary classification and regression with appropriate outputs and loss functions.

12. Training on Real Data

End-to-end pipeline covering loading, preprocessing, batching and model training.

13. Deployment and Reuse

Parameter serialization, model loading and inference on unseen data.